

Rima Fathallah Software Engineering Graduate Student

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SUMMARY

A Computer Science graduate specialized in AI-driven solutions and intelligent systems, with over 3 years of practical experience in **Python**, **machine learning**, and **work automation**. Skilled in developing data-driven applications, optimizing analytical processes, and integrating intelligent models into real-world systems. Currently seeking an opportunity to advance my research capabilities and strengthen my professional expertise in AI and data science.

SKILLS

Programming Languages:

Python, R, C++, JavaScript, C#, SQL

Machine Learning & Data Science:

Scikit-learn, XGBoost, Random Forest, Pandas, NumPy, SciPy, Feature Engineering, Model Evaluation

Computer Vision & Research:

OpenCV, Image Processing, Object Detection, Image Classification, Experimental Design, Ablation Studies, Reproducible Research

Development & Tools:

Git, Docker, Flask, REST APIs, ReactJS, MySQL, PostgreSQL, LaTeX, Linux

Deep Learning & Generative AI:

PyTorch, TensorFlow, CNNs, RNNs/LSTMs, Transformers, Hugging Face, LLMs, Fine-tuning, AI Agents

PROFESSIONAL EXPERIENCE

AI Developer Engineer (Full Time) 🌐, Exceed Your Limits Marketing

Mar 2026 – Present | Germany, Remote

- Develop an AI-powered website that transforms user prompts into functional websites and automates code generation.
- Develop and optimize course, webinar, and payment workflows; integrate REST APIs and DB systems with 100K+ records.

Technologies: Python, AI/LLMs, C#, .NET, React, TypeScript, REST APIs, SQL, Docker, Git/GitLab

Product: <https://eyl-educate.com/> 🌐

Researcher (MITACS GRI'25 Intern) 🌐, Concordia University

Jan 2025 – Jul 2025 | Montreal, Canada

- Developed, calibrated, and validated ML models and prepared and analyzed datasets to support the research.
- Designed an automated platform, prototyped solutions, evaluated results, and documented findings for research teams.

Technologies: Python, Machine Learning, Data Analysis, Pandas, Scikit-learn, SQL, Jupyter, Git, XGBoost, LSTM

Junior Instructor (Part-Time) 🌐, Go My Code

Sep 2023 – Apr 2025 | Sousse, Tunisia

- Mentored students across AI, game development, and creative technologies, achieving a 95% project completion rate.
- Delivered hands-on training in Generative AI, Unity, Scratch, and Adobe, improving student performance by 30%.

Technologies: Generative AI, Unity, Scratch, Adobe Illustrator

Junior Artificial Intelligence Engineer (Internship) 🌐, Ozeol.com

Jun 2024 – Sep 2024 | Sousse, Tunisia

- Designed optimized data pipelines, improving processing efficiency by 27% and analyzing 5K+ records for realtime reports.
- Built data collection and validation workflows that reduced data inconsistencies by 13% while supporting secure platforms.

Technologies: Python, FastAPI, SQL, Pandas, Web Scraping, Data Analysis, Data Pipelines, Real-Time Reporting

Junior Software Engineer (Internship) 🌐, Draexlmaier

Aug 2023 – Oct 2023 | Sousse, Tunisia

- Designed an anonymous employee feedback platform that increased employee engagement by 25%.
- Built real-time dashboards and analytics that improved management's response to workplace issues by 20%.

Technologies: Power Apps, Power BI, Data Analytics, Data Visualization, SQL

Automation Engineer (Part-Time), Elit school of languages Sousse

Jun 2022 – Oct 2022 | Sousse, Tunisia

- Automated data processing and management workflows, reducing processing time and improving productivity by 30%.
- Developed an automated certification system with real-time data updates, improving user satisfaction by 20%.

Technologies: Python, Pandas, NumPy, Data Automation, Data Processing, Real-Time Systems

Junior Web Developer (Internship) 🌐, BillCom Consulting

Jul 2022 – Sep 2022 | Sousse, Tunisia

- Designed and optimized a scalable data platform, improving database performance by 18%.
- Automated Linux-based deployment and maintenance workflows, improving platform stability and reducing manual task.

Technologies: Python, MySQL, SQL, Linux, Database Optimization, Data Engineering, Automation

CERTIFICATES

Fundamentals of Deep Learning 🌐

NVIDIA Deep Learning Institute

Building Transformer-Based Natural Language Processing (NLP and LLM) Applications 🌐

NVIDIA Deep Learning Institute

Applications of AI for Predictive Maintenance 🌐

NVIDIA Deep Learning Institute

AI-900: Azure AI Fundamentals 🌐

Microsoft Certified

Introduction to Security Principles in Cloud Computing 🌐

Google Cloud Security - Coursera

RESEARCH AND DEVELOPMENT

AI-Driven Modeling of Neuroplasticity Trajectories in Pediatric Cerebral Palsy Aug 2026 – Present
Leading an ongoing AI research project in collaboration with Sahloul University Hospital and Hached University Hospital.

PROJECTS

- NEXOVA** *🔗, Startup Idea* Sep 2024 – present
AI-powered platform that analyzes and optimizes energy consumption to reduce electricity usage and improve efficiency.
Technologies: Python, Machine Learning, Data Analysis, Pandas, NumPy, Scikit-learn
- Autonomous Data Analysis Agent**, *Personal Development Project* Jul 2026 – Sep 2026
Developed an AI agent that autonomously cleans, analyzes, visualizes, detects anomalies, and reports based on requests.
Technologies: Python, Pandas, NumPy, Scikit-learn, LLMs/SLMs, Function Calling, LangGraph, FastAPI, Plotly, Docker
- Multimodal AI System for Patient Risk Assessment**, *Mid-Term Project* Apr 2026 – Jun 2026
Developed a multimodal AI system combining clinical data, medical images, and patient history to predict patient risk.
Technologies: Python, CNN, Vision Transformers (ViT), XGBoost, Transformers, SHAP, Grad-CAM, Pandas, Scikit-learn
- Explainable Multilingual Misinformation Detection**, *Evaluation Project* Feb 2026 – Apr 2026
Developed a multilingual NLP system for misinformation detection across languages.
Technologies: Python, NLP, TF-IDF, LSTM, BERT, AraBERT, Multilingual Transformers, Explainable AI, Scikit-learn, Hugging Face
- ToolMind**, *School Project* Dec 2025 – Feb 2026
Developed an intelligent AI agent that plans tasks, selects tools/APIs, and generates responses using structured tool routing.
Technologies: Python, SLMs (Llama), Hugging Face, LangGraph, LangChain, FastAPI, REST APIs, Scikit-learn, Docker, MLflow
- DEXTRA**, *Congress Challenge* Sep 2025 – Dec 2025
AI-powered smart gloves enabling real-time bidirectional translation between sign language and speech using embedded ML.
Technologies: Python, Machine Learning, Embedded AI, Computer Vision, Sensor Data, IoT
- Talynk** *🔗, Personal Development Project* Aug 2025 – Nov 2025
Developed an AI-powered platform for CV generation and interview preparation using NLP and machine learning.
Technologies: Python, NLP, Machine Learning, LLMs, Text Processing, Scikit-learn

EDUCATION

- Engineering Degree in Data Science**, *Ecole Polytechnique de Sousse - Evening courses* *🔗* Sep 2025 – Present | Sousse, Tunisia
Main Modules: MLOps, AIoT, MERN, NLP, C, AI, CyberSec
- Bachelor's degree in Computer Science (Licence)** Sep 2023 – Jul 2025 | Sousse, Tunisia
École supérieure des sciences et de la technologie de Hammam Sousse(ESSTHS) *🔗*
Main Modules: Algorithms, Industrial Programming, Advanced OO Programming, ISTQB, web programming, IoT, AI & ML
- Bachelor's degree in Mathematics (Licence)** Sep 2021 – Jun 2023 | Sousse, Tunisia
École supérieure des sciences et de la technologie de Hammam Sousse(ESSTHS) *🔗*
Main Modules: Statistics with R, Probabilities, Operational research, Queuing Theory, Algebra & Analysis

LANGUAGES

English	French	German	Arabic
DET - C1	DELFI - B2	ELIT - A1	Native

AWARDS

- AI NIGHT Challenge** *🔗, ARSII, Second Place in the 2026 Edition, Fourth place in the 2024 Edition* 05 May 2026
- Herotopia Challenge** *🔗, IEEE TSYP 13, First Place* 24 Dec 2025

VOLUNTEERING

- UNESCO Chaire EDE X IEEE**, *Program Co-Lead* Jan 2026 – Dec 2026
- Led 30+ ambassadors to develop monthly sustainable projects, delivering measurable impact across Tunisia.
- IEEE Tunisia Section - SIGHT Group**, *Project Coordinator* Jan 2025 – Dec 2026
- Coordinates several Sustainable Projects that align with SDGs in Tunisia, Sustainable Technology, ThyroCare, Bled el Khir...
- IEEEExtreme 18.0 - Region 8** *🔗, Co-Lead* Sep 2023 – Oct 2024
- Led a team of 90+members across Africa, Europe, and ME to create world's largest CP competition with 10K+ participants.
- IEEE Tunisia Section - ESSTHS Student Branch** *🔗, Chair* Jan 2023 – Dec 2023
- Led the largest SB in Tunisia. winning 1st place in 8+ competitions. Organized TSYP 11 with 1,200+ participants and 15 challenges.

POSITIONS

UNESCO Chaire EDE X IEEE, Program Co-Lead	Jan 2026 - Dec 2026
Led 30+ ambassadors to develop monthly sustainable projects, delivering measurable impact across Tunisia.	
IEEE Tunisia Section - SIGHT Group, Project Coordinator	Jan 2025 - Dec 2026
Coordinates sustainable projects aligned with the SDGs in Tunisia, including Sustainable Technology, ThyroCare, and Bled el Khir.	
IEEEExtreme 18.0 - Region 8, Co-Lead	Sep 2023 - Oct 2024
Led a team of 90+ members across Africa, Europe, and the Middle East for a programming competition with 10K+ participants.	
IEEE Tunisia Section - ESSTHS Student Branch, Chair	Jan 2023 - Dec 2023
Led the largest student branch in Tunisia, with first place in 8+ competitions. Organized TSYP 11 with 1,200+ participants and 15 challenges.	

CONGRESSES

IEEE TSYP 13	24 Dec 2025
Attended. First place in the Herotopia Challenge.	
IEEE TSYP 11	2023
Organized the congress as Chair of the ESSTHS student branch, with 1,200+ participants and 15 challenges.	
DEXTRA, Congress Challenge	Sep 2025 - Dec 2025
AI-powered smart gloves enabling real-time translation between sign language and speech using embedded machine learning.	

IEEE MENTORING

AI	IEEE mentor
Mentored an IEEE session on AI.	
NLP	IEEE mentor
Mentored an IEEE session on natural language processing.	
Explainable AI	IEEE mentor
Mentored an IEEE session on explainable AI.	